WasteWise

Smart Waste Management System - Project Overview

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1. Project Overview

WasteWise is a full-stack DBMS Mini Project designed to optimize municipal garbage collection. Through dynamic tracking, smart bin fill level calculations, real-time alert notifications, and automated scheduling workflows, the system connects local citizens, collection workers, and municipal administrators onto a unified administrative portal.

2. Technology Stack & Component Roles

Frontend: React.js, Vite, TailwindCSS, Chart.js
Role: Acts as the presentation layer. It manages routing, global state, authentication contexts, and renders responsive user-specific dashboards for citizens, workers, and administrators. Includes interactive visual graphs (such as average waste level by area and weekly collection statistics).

Backend: Node.js, Express.js
Role: Acts as the application logic layer. It parses requests, handles JWT authorization middleware, applies data validations, and interacts with MongoDB via Mongoose. It exposes secure endpoints for CRUD operations across all system collections.

Database: MongoDB Atlas (Mongoose ORM)
Role: Acts as the data persistence layer. Uses a cloud document-oriented model with schema validations to store entities like Users, Smart Bins, Tasks, Reports, Vehicles, Notifications, Feedbacks, and Activity logs. It supports relational integrity audits (like cascading assignments and log history creation) in controllers.

3. Key Portal Roles & Features

A. Administrator Portal (Admin Dashboard)

- Bin Management: Register smart bins, assign locations, update parameters, and monitor fill capacities.
- Worker & Fleet Allocation: Assign collection duties to workers and allocate garbage trucks/vehicles.
- Activity Logs & Database Explorer: Direct dashboard UI component to read system logs and browse raw MongoDB collections.

B. Collection Worker Portal

- Task Schedule: Access assigned collection duties with detail cards on bin locations.
- Capacity Alerting: Automated system warnings flag managed bins whose fill level crosses or equals 80%.
- Mark Completion: Update task progress in real time to trigger status changes on linked citizen complaints.

C. Citizen Portal

- Waste Complaint Form: Report bin overflow or garbage issues with location coordinates and details.
- Progress Tracking: Keep track of submitted complaints and receive notifications when tasks are assigned or resolved.
- Feedback & Reviews: Submit community feedback to improve civic collection services.

4. Database Schema Relationships & Integrity

- Bin is mapped to a specific Zone/Area and can have an assigned Worker.
- Waste Report is submitted by a Citizen and holds fields for Bin, Status, and Location.
- Collection Task is created by an Admin to clean a Bin, linking a specific Worker, Vehicle, and collection date.
- Activity Log audits all actions (e.g. 'Bin Created', 'Task Assigned') capturing the Actor, Entity Type, and timestamp.
- Notification maps recipientId to user targets to deliver dynamic alerts on status transitions.